

CSIS1614 Theory Consolidation Set 1

Mixed define / compare / explain questions based on LU1-LU6.

Answer in your own words, like you would in a test.

Methods (LU5)

1. What is a method?
2. What is the difference between a **void method** and a **value-returning method**?
3. Explain the difference between a **parameter** and an **argument**.
4. Why must a value-returning method use **return**?
5. Explain the difference between a value parameter, a **ref** parameter, and an **out** parameter.
6. Why must **ref** appear in both the method header and the method call?
7. What is a default/optional parameter?
8. What is the purpose of named parameters?

Loops (LU4)

9. Why do programs use loops?
10. Name and explain the 4 main components of a loop.
11. What is the difference between a **pre-test loop** and a **post-test loop**?
12. Compare **while**, **for**, and **do...while**. State when each is useful.
13. What is a **counter-controlled** loop?
14. What is a **sentinel-controlled** loop?
15. What is a **state-controlled** loop?
16. Explain the difference between **break** and **continue**.
17. What is a nested loop?
18. Why does the inner loop restart every time the outer loop runs?

Making Decisions (LU3)

19. What is a Boolean expression?
20. Explain the difference between equality operators, relational operators, and logical operators.
21. What is the difference between a one-way **if** and a two-way **if...else**?
22. What is a nested **if** statement?
23. What is the purpose of a **switch** statement?
24. When does the **default** block of a **switch** run?

- 25. What does the ternary operator **?:** do?
- 26. Explain short-circuit evaluation for **&&** and **||**.
- 27. What is variable scope?

Arrays and Strings (LU6)

- 28. What is an array?
- 29. What is an array element?
- 30. What is an index/subscript?
- 31. Why are arrays called **zero-based**?
- 32. Explain the difference between declaration, instantiation, and initialization for arrays.
- 33. Why is **array.Length - 1** the last valid index?
- 34. What is the difference between traversing an array with **for** and **foreach**?
- 35. Why can **foreach** read array values but not assign into array elements?
- 36. Why should **for** usually use **< array.Length** instead of **<= array.Length**?
- 37. How are arrays used as method parameters?
- 38. Why do changes to array elements inside a method affect the original array?
- 39. Why does array assignment with **=** not create a separate copy of the array elements?
- 40. What does the **params** modifier do?
- 41. Why must a **params** parameter be the last parameter in the method header?
- 42. Why can a string be treated like a collection?
- 43. What does it mean that strings are immutable?
- 44. Explain what each string method does: **Compare()**, **IndexOf()**, **Substring()**, **Remove()**, **Split()**, **Insert()**.

Broader programming theory (LU1-LU2)

- 45. What is a program?
- 46. What is software?
- 47. What is the difference between **system software** and **application software**?
- 48. What is a compiler?
- 49. What is the input-processing-output structure?
- 50. Explain the steps of the software development process in the correct order.